

EDUCATION

- **The University of Chicago, Booth School of Business** Chicago, IL
Ph.D. student in Management Science / Operations Management Sep.2025 –
- **The University of Chicago** Chicago, IL
Master of Science in Computational and Applied Math Sep.2024 – Jun.2025
- **Hong Kong University of Science and Technology** Hong Kong SAR
Bachelor of Science in (Pure) Mathematics (First Class Honors) Sep.2020 – Jun.2024
- **Columbia University in the City of New York** New York City, NY
Exchange Student at Columbia College Jan.2023 – May.2023

PUBLICATION

- Haoyu Wu, Ao Tang, “Dynamic Competition in Elections: A Noncooperative Game of Candidate Positioning”, *Conference on Decision and Control*, 2026 forthcoming.
- Haoyu Wu, Ao Tang, “An Even Tighter Bound for the Shapley-Folkman-Starr Theorem”, *Journal of Mathematical Economics*, V114, 2024. doi.org/10.1016/j.jmateco.2024.103028.

RESEARCH PROJECTS

- **Optimal Transport for a Multiclass $G/GI/N + GI$ Queue (Working)**
Keywords: Queueing Scheduling, Optimal Transport with Prof. Bahar Taskesen & Prof. Amy Ward
 - **Optimal Transport Approach:** We propose a new approach to the multiclass many-server queueing scheduling problem in the fluid limit, inspired by optimal transport theory. The reformulation inspired by optimal transport provides the optimal scheduling policy under a general discipline scheduling policy class in a finite horizon.
 - **Talks about the paper:** MOPTA 2026, Informs Annual Meeting 2026 TE30
- **Deep Reinforcement Learning for Integer-valued Action Space (Working)**
Keywords: Deep Reinforcement Learning, Autoregression with Dr. Ayoub Foussoul & Prof. Ozan Candogan
 - **Sequential Sampling:** We are developing a DRL approach to an integer-valued action space with a sequential action construction.
- **Dynamical Monotone System in Competitive Equilibrium Convergence (Working)**
Keywords: General Equilibrium, Supermodular Game with Prof. A. Kevin Tang
 - Building on the technical fluid-system results for dynamic monotone systems, we seek to provide an economic interpretation of the convergence conditions in a pure exchange economy.
- **Dynamic Competition in Elections: A Noncooperative Game of Candidate Positioning**
Keywords: Dynamic Game ; Conference on Decision and Control 2026 forthcoming with Prof. A. Kevin Tang
 - **Abstract:** A noncooperative dynamical model of elections is developed. We analyze the case of two candidates competing for a fixed, stationary electorate over a finite horizon. Numerical examples illustrate how voter distribution, campaign duration, and initial positions shape equilibrium candidate trajectories. We extend the model to incorporate credibility losses from policy repositioning.
- **An Even Tighter Bound for the Shapley-Folkman-Starr Theorem**
Journal of Mathematical Economics, V114, 2024 with Prof. A. Kevin Tang
 - **Abstract:** Based on a previous refined Shapley-Folkman lemma, we derive a tighter error bound for the Shapley-Folkman-Starr theorem and apply the result to the course allocation problem.
 - **Main Idea:** Instead of the traditional method of separately estimating each component of the Minkowski sum, we considered all components jointly and utilized the constraint on convexity from the refined Shapley-Folkman lemma to reach a tighter bound.

CONFERENCE TALK & POSTER PRESENTATION

- Chicago Operations Workshop (Poster)
- Stochastic Network Conference (Poster)
- MOPTA 2026
- INFORMS Annual Meeting TE30

Chciago, 2026 June
Chicago, 2026 June
Lehigh University, 2026 August
San Francisco, 2026 November

UNDERGRAD RESEARCH STUDIES

- **Multi-structured Signal Recovery Methods** Jul. 2022 - Dec. 2022
Keywords: Signal Processing, Inverse Problem, Convex Opt. *Advisor: Prof. Jian-Feng Cai*
 - **Linear Combinations Approach:** Investigated recovery methods for multi-structured signals, such as sparse and low-rank matrices and tensors. Evaluated the optimality of linear combinations of recovery norms and assessed the effectiveness of the method of the square norm for low-rank tensors.
 - **Atomic Norms Approach:** Identified the atomic norm as a recovery method for multi-structured signals. Explored the effectiveness of a new atomic norm and its corresponding atomic set, constructed from the intersection of known atomic sets for different structures.
- **Computational Optimal Transport** Feb. 2022 - Jun. 2022
Keywords: Entropy, Sinkhorn Algorithm, Convex Opt. *Advisor: Prof. Jian-Feng Cai*
 - **Entropic Regularization:** Investigated the entropic regularization and extensively examined the local and global convergence properties of the Sinkhorn algorithm.
 - **Sinkhorn Algorithm:** Explored the basic properties of the Sinkhorn algorithm. Implemented modifications to the Sinkhorn algorithm, such as the Greenhorn algorithm, with the aim of acceleration.
- **Shrinking a Convex Curve via Heat Equation** Jun. 2021 - Aug. 2021
Keywords: Flow Geometry, Heat equation *Advisor: Prof. Tsz-Ho Fong*
 - **Rigorous Analysis:** Formal analysis of shrinking a convex curve to a point via the heat equation.

AWARDS

- Doctoral Centennial Fellowship 2025 Fall
- Lee's Pharmaceutical - Kanya Lee Scholarship 2023 Fall
- University's Scholarship Scheme for Continuing UG Students (Ranked 10%) & Dean's List 2020-2024
- Chern Class Talent Scholarship (For Math Students Pursuing Challenging Courses) 2020-2024
- Honorable Mention Award, Math Department, HKUST 2023

TEACHING ACTIVITIES

- Grader of Honor Calculus I, HKUST 2023 Fall
- Teaching Assistant at Math Support Center, HKUST 2022 Fall